

WEN-HORNG SHEU

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Professional Summary: Ph.D. candidate in Computer Science specializing in algorithms for large-scale and memory-constrained computation. Over five years of research experience and a strong background in competitive programming.

EDUCATION

Ph.D. in Computer Science, University of California, Davis; GPA: 4.0/4.0 2023-Present

- Expected Graduation: June 2028 (06/28)
- Research Area: Distributed Algorithms, Streaming Algorithms

M.S. in Computer Science, National Tsing Hua University; GPA: 3.9/4.0 2019-2021

- Selected Coursework: Approximation Algorithms, Computational Geometry

B.S. in Computer Science, National Tsing Hua University; GPA: 3.85/4.0 2015-2019

- Selected Coursework: Advanced Data Structures, Randomized Algorithms, Parallel Algorithm Design

SELECTED PUBLICATIONS

Faster Semi-streaming Matchings via Alternating Trees, ICALP '25

- Designed an algorithm for computing an approximate matching in memory-constrained settings.
- Improved a STOC '22 result, reducing pass complexity from $O(\epsilon^{-19})$ to $O(\epsilon^{-6})$ via a new data structure.

A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic, SPAA '25

- Designed a framework for $(1 + \epsilon)$ -approximate matching, improving results from FOCS '24 and SODA '25.
- In distributed settings, reduced round-complexity dependence on ϵ from $\Omega(\epsilon^{-52})$ to $O(\epsilon^{-7} \log(1/\epsilon))$.
- In dynamic settings, reduced time-complexity dependence on ϵ from $\exp(1/\epsilon)$ to $\text{poly}(1/\epsilon)$.

Faster MPC Algorithms for Approximate Allocation and Matching in Uniformly Sparse Graphs, SPAA '25

- Developed a new distributed algorithm with applications in online advertising and load balancing.
- Reduced round complexity from $O(\log n)$ to $O(\sqrt{\log n})$ using random sampling, extending an ICML '18 result.

SELECTED HONORS

- **Shadow Program Committee Member** for ACM SPAA 2026
 - Invited to serve on the shadow program committee for a leading conference in parallel algorithms.
 - Reviewed 6 submissions; participated in committee discussions and contributed to paper evaluations.
- **Gold Award** in the 2019 ICPC Asia Pacific Taipei-Hsinchu Regional Contest
 - Competed in 6+ international programming contests, including multiple ICPC regional contests.
 - Solved 1,500+ algorithmic programming problems across graph algorithms and data structures.
- **Grandmaster on Codeforces**
 - Placed in the top 1% among 10,000+ global contestants in four Codeforces programming contests.
 - Ranked as a Grandmaster, within the top 1% globally on Codeforces as of February 2021.

SKILLS

Programming Languages

C++, Python

Tools

Git, L^AT_EX

Languages

English (fluent, TOEFL 105/120), Chinese (native)